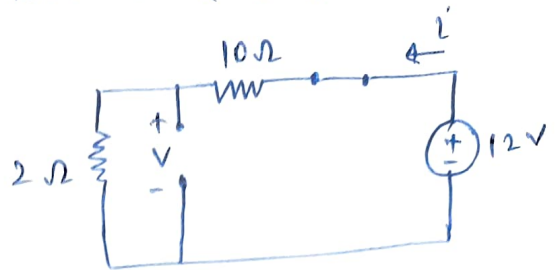


1. a) Draw the circuit before $t=0$; i.e. switch open
 The steady state circuit will look like

so,

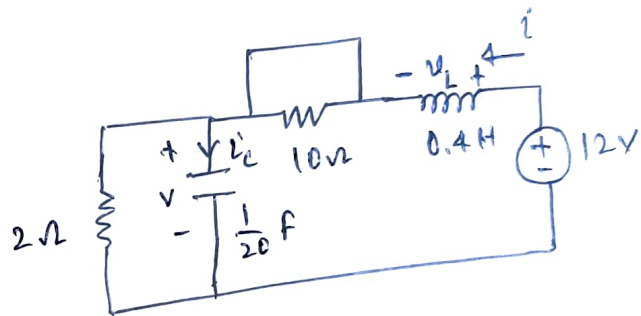
$$i(0^+) = i(0^-) = \frac{12}{2+10} = 1 \text{ A}$$

$$v(0^+) = v(0^-) = 2 \times 1 = 2 \text{ V}$$



- b) Draw the circuit at $t=0$; i.e. when switch is closed

$\Rightarrow$ no current will flow through 10Ω resistor as it is short circuited



so, apply KVL

$$12 = v_L + v$$

$$12 = L \frac{di}{dt} + v$$

at $t=0^+$

$$12 = L \frac{di(0^+)}{dt} + v(0^+)$$

$$12 = L \frac{di(0^+)}{dt} + 2$$

$\left[\because v(0^+) = 2 \text{ V from (a)} \right]$

$$\frac{di(0^+)}{dt} = 25 \text{ A/s}$$

Now,

apply nodal analysis at node v

$$i = i_c + \frac{v}{2}$$

at $t = 0^+$

$$i(0^+) = i_c(0^+) + \frac{v(0^+)}{2}$$

$$1 = C \frac{dv(0^+)}{dt} + \frac{2}{2}$$

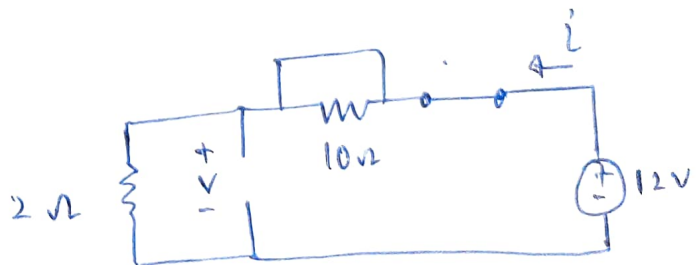
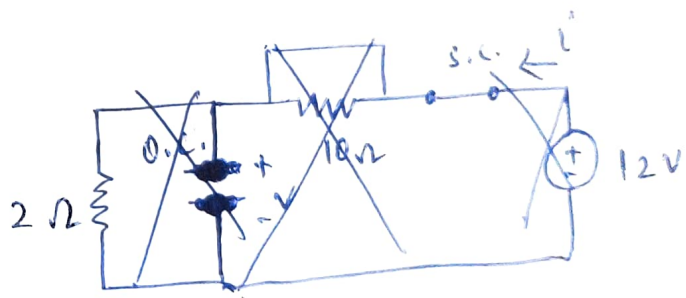
$$\left[i(0^+) = 1A, v(0^+) = 2V \text{ from (a)} \right]$$

$$\boxed{\frac{dv(0^+)}{dt} = 0}$$

(c) Draw the circuit in steady state after switch is closed at $t=0$

$$i(\infty) = \frac{12}{2} = 6A$$

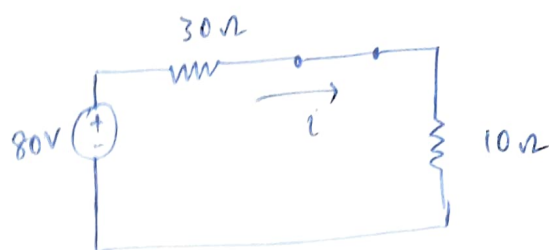
$$v(\infty) = 2 \times 6 = 12V$$



2. Ans: Step 1: Draw the circuit before $t=0$ and find out the initial current & voltage value

$$i(0^-) = \frac{80}{30+10} \text{ A} = 2 \text{ A}$$

$$v(0^-) = 0 \text{ V (given)}$$

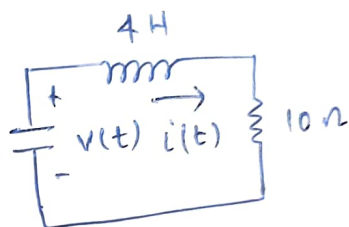


$$\text{so, } i(0^-) = i(0^+) = 2 \text{ A}$$

$$v(0^-) = v(0^+) = 0 \text{ V}$$

Step 2 Draw the circuit after switch operation i.e. $t > 0$

Now, At $t > 0$, The circuit is source free second order ^{series} network



$$\text{so, } \alpha = \frac{R}{2L} = \frac{10}{8} = 1.25$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times 0.25}} = 1$$

as $\alpha > \omega_0$, the response will be overdamped

so, for the source free second order series network, the current equation can be written as

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{--- (1)}$$

$$s_1, s_2 = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -2, -0.5$$

$$\text{so, (1) } \Rightarrow i(t) = A_1 e^{-2t} + A_2 e^{-0.5t} \quad \text{--- (2)}$$

Step 3 Use the initial conditions to find out the constants (A_1, A_2) value

so, at $t=0$, $i(0) = 2 \text{ A}$

so, (2) $\Rightarrow$ $\boxed{A_1 + A_2 = 2}$ — (3)

and at $t=0$, $v(0) = 0 \text{ V}$

so, from step 2, KVL equation can be written as

$$v(t) = L \frac{di(t)}{dt} + R i(t)$$

$$v(0) = L \frac{di(0)}{dt} + R i(0)$$

$$0 = 4 \frac{di(0)}{dt} + 20$$

$$\boxed{\frac{di(0)}{dt} = -5}$$

Now, Differentiate Eqⁿ (2) and put $t=0$, we get

$$\boxed{2A_1 + 0.5A_2 = 5}$$
 — (4)

solve Eqⁿ (4) & (3), $\boxed{A_1 = 8/3}$, $\boxed{A_2 = -2/3}$

so, from step 2, $v(t) = -\frac{1}{c} \int i dt$ — (5)

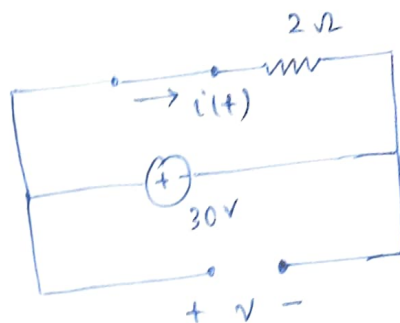
and, $i(t) = \frac{8}{3} e^{-2t} - \frac{2}{3} e^{-0.5t}$ (from equation (2))

so, eqn (5) $\Rightarrow$ $\boxed{v(t) = \frac{16}{3} [e^{-2t} - e^{-0.5t}]; t > 0}$

Q4) Ans: Step 1 :- Draw the circuit before $t = 0$ (steady state)

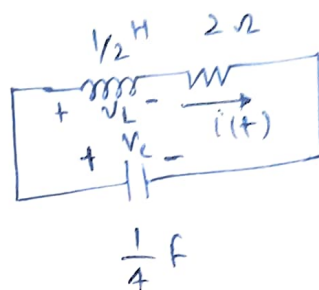
$$i(0^-) = i(0^+) = \frac{30}{2} = 15 \text{ A}$$

$$v(0^-) = v(0^+) = 30 \text{ V}$$



Step 2 Draw the circuit after $t = 0$

Now, this is source free
Second order series circuit



$$\alpha = \frac{R}{2L} = 2$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{8}$$

so, $\alpha < \omega_0 \Rightarrow$ response will be underdamped

$$\text{so, } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = 2$$

Now, for underdamped response,

$$i(t) = e^{-\alpha t} [A_1 \cos \omega_d t + A_2 \sin \omega_d t] \quad \text{--- (1)}$$

$$\text{so, } i(t) = e^{-2t} [A_1 \cos 2t + A_2 \sin 2t] \quad \text{--- (2)}$$

Step 3 :- Use initial conditions to find out A_1 & A_2 value

so, from step 1, at $t=0$, $i = 15 \text{ A}$

so, from eqⁿ (2) $\Rightarrow$ $\boxed{A_1 = 15}$

Also, Write KVL eqⁿ from step 2

$$V_L = V_R + iR \quad \text{--- (3)}$$

at $t=0$, $i(0) = 15 \text{ A}$, $V_L(0) = 30 \text{ V}$

so, (3) $\Rightarrow V_L = 0$

$$L \frac{di}{dt} = 0$$

$$\boxed{\frac{di}{dt} = 0} \quad \text{--- (4)}$$

so, Differentiate eqⁿ (2) to find out $\frac{di}{dt}$ & put

$$\frac{di}{dt} = 0 \quad \text{from eqⁿ (4), we get}$$

$$A_1 - A_2 = 0$$

so, $\boxed{A_2 = A_1 = 15}$

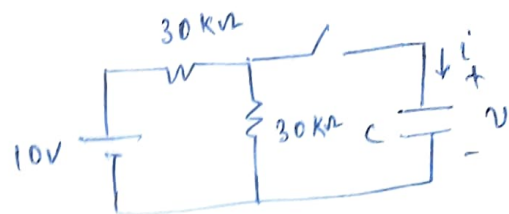
so, finally from eqⁿ (2)

$$\boxed{i(t) = 15e^{-2t} [\cos 2t + \sin 2t] ; t > 0}$$

⑤ step 1 Draw the circuit before $t = 0$

$$i(0^-) = i(0^+) = 0$$

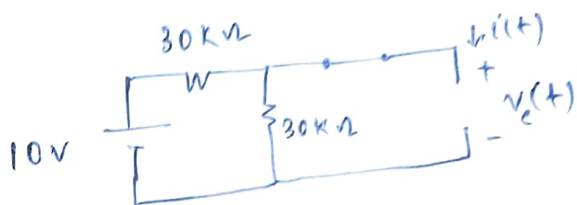
$$v(0^-) = v(0^+) = 0$$



step 2

Draw the circuit when switch is closed at $t = 0$

for $t > 0$ (at steady state)



$$v(\infty) = 5V$$

step 3

write down voltage expression across capacitor

$$v_c(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau} \quad \text{--- (1)}$$

$$\tau = RC = 3 \times 10^{-6} \times 15 \times 10^3 = 45 \times 10^{-3} s$$

$$\text{so, (1) } \Rightarrow v_c(t) = 5 [1 - e^{-22.22t}] \quad V ; t > 0$$

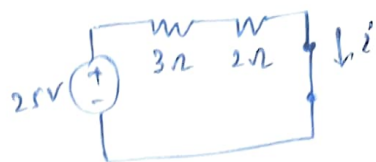
$$i(t) = C \frac{dv}{dt}$$

$$= 3 \times 10^{-6} \times 111.11 e^{-22.22t}$$

$$\text{so, } i(t) = 333.33 e^{-22.22t} \mu A ; t > 0$$

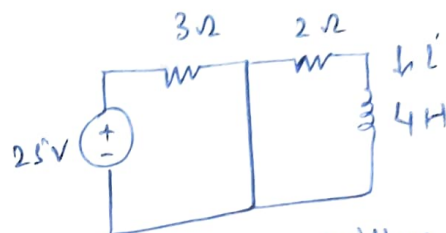
6. ① Draw the circuit before switch gets closed i.e. $t < 0$

$$i(0) = \frac{25}{3+2} = 5 \text{ A}$$



Now, Draw the circuit after switch gets closed i.e. $t > 0$

$$i(\infty) = 0 \text{ A}$$



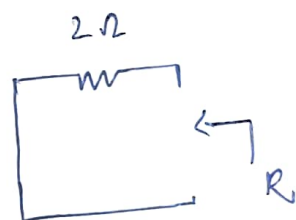
so, current expression through inductor can be written as

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\tau} \quad \text{--- (i)}$$

where $\tau = L/R$

so, $\tau = \frac{4}{2}$

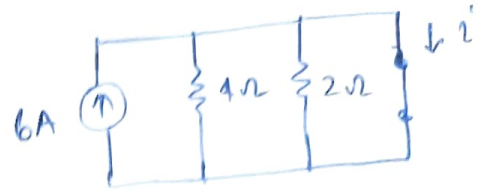
$$\tau = 4/2 = 2 \text{ sec}$$



so, ① $\Rightarrow$
$$i(t) = 5e^{-0.5t} \text{ A}, \quad t > 0$$

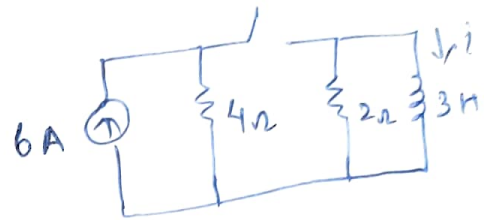
6. (b) or step 1 :- Draw the circuit before $t=0$; i.e. when switch was closed

$$i(0) = 6 \text{ A}$$



step 2 :- Draw the circuit when switch gets opened i.e. after $t=0$

$$i(\infty) = 0 \text{ A}$$



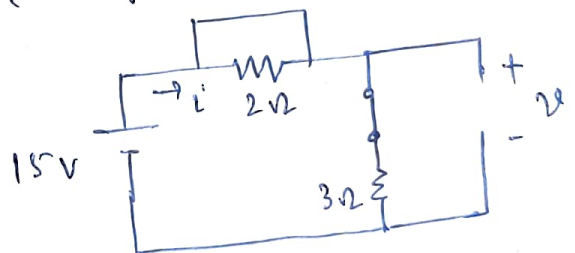
$$\text{so, } i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$i(t) = 6e^{-2/3t} \text{ A ; } t > 0$$

7. (a) Ans. Draw the circuit when both switches are closed for a long time (steady state)

$$i(0) = \frac{15}{3} = 5 \text{ A}$$

$$v(0) = 15 \text{ V}$$



so, Energy will be stored in inductor & capacitor

$$E_L = \frac{1}{2} Li^2 = \frac{1}{2} \times 1 \times 5^2 = 12.5 \text{ J}$$

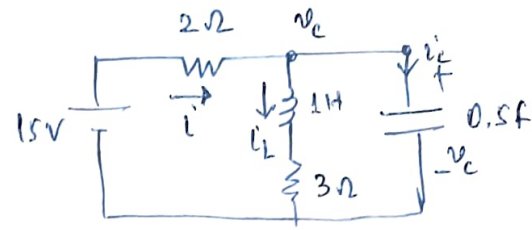
$$E_C = \frac{1}{2} Cv^2 = \frac{1}{2} \times 0.5 \times 15^2 = 56.25 \text{ J}$$

7. (b)

Draw the circuit when s_2 is opened at $t = 0$

$$i_L(0^+) = i_L(0^-) = 5 \text{ A}$$

$$v_c(0^+) = v_c(0^-) = 15 \text{ V}$$



Apply nodal analysis at node v_c

$$+i = i_L + i_c$$

$$\text{at } t = 0^+$$

$$i(0^+) = i_L(0^+) + i_c(0^+) \quad \text{--- (1)}$$

so, write KVL eqn

$$15 = 2i + v_c$$

$$\text{at } t = 0^+$$

$$15 = 2i(0^+) + v_c(0^+)$$

$$15 = 2i(0^+) + 15$$

$$\text{so, } \boxed{i(0^+) = 0 \text{ A}}$$

$$\text{so, (1) } \Rightarrow \boxed{i_c(0^+) = -i_L(0^+) = -5 \text{ A}}$$

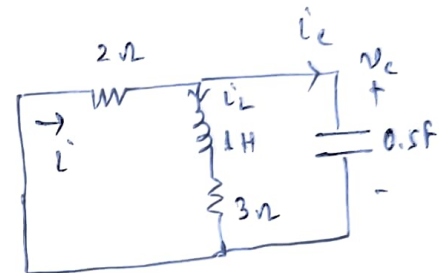
7. c) Since the given circuit is not a standard RC, RL or RLC (series/parallel) form, we need to derive the solution for it. The order of the circuit is 2.

So, complete response is sum of forced response + natural response.

Let us find natural response first with s_1 closed and s_2 open

$$\text{KCL: } i = i_L + i_C \quad \text{--- (1)}$$

$$\text{KVL: } 2i + 3i_L + 1 \cdot \frac{di_L}{dt} = 0 \quad \text{--- (2)}$$



$$i = -\frac{v_C}{2} \quad \text{--- (3)}$$

$$\text{(1)} \Rightarrow i_L = i - i_C$$

$$i_L = i - C \frac{dv_C}{dt}$$

$$i_L = i + 2C \frac{di}{dt} \quad \text{--- (4)} \quad [\text{from eqn (3)}]$$

$$\text{So, eqn (2)} \Rightarrow \frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 5i = 0$$

$$\text{characteristics eqn: } s^2 I + 4sI + 5I = 0$$

$$I(s^2 + 4s + 5) = 0$$

$$s_1, s_2 = \frac{-4 \pm \sqrt{16 - 20}}{2}$$

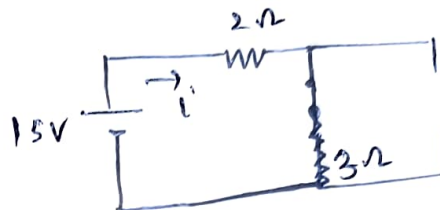
$$s_1, s_2 = -2 \pm j \quad \rightarrow \text{Underdamped response}$$

So, template solution of 2nd order source free circuit is

$$\begin{aligned} i_{\text{natural}}(t) &= e^{-\alpha t} [A_1 \cos \omega_d t + A_2 \sin \omega_d t] \\ i_{\text{natural}}(t) &= e^{-2t} [A_2 \sin t + A_1 \cos t] \end{aligned} \quad \text{--- (5)}$$

Now, find out the forced response

$$i_{\text{forced}}(t) = \frac{15}{2+3} = 3 \text{ A}$$



So, complete response

$$i(t) = 3 + e^{-2t} [A_1 \cos t + A_2 \sin t] \quad \text{--- (6)}$$

Now at $t = 0^+$, $i(0^+) = 0 \text{ A}$ [from (b)]

So,

$$A_1 = -3$$

Also, at $t = 0^+$, $i_L(0^+) = 5 \text{ A}$

So, eqⁿ (4) $\Rightarrow$

$$A_2 = -1$$

So,

$$i(t) = 3 - 3e^{-2t} \cos t - e^{-2t} \sin t$$

Also,

$i(t)$ can be written as for underdamped ~~res~~ natural response $\therefore$

$$i(t) = Ae^{-\alpha t} [\sin \omega_d t + \theta]$$

so, complete response

$$i(t) = 3 + Ae^{-2t} [\sin t + \theta] \quad \text{--- (A)}$$

$$\text{at } t=0, \quad i = 0$$

$$\text{so, (A) } \Rightarrow \boxed{A \sin \theta = -3} \quad \text{--- (B)}$$

$$\text{and at } t=0, \quad i_L = 5 \text{ A}$$

$$\text{so, } i_L = i + \frac{di}{dt}$$

$$\text{on simplifying, } \boxed{A \cos \theta = -1} \quad \text{--- (C)}$$

$$\frac{\text{Eqn (B)}}{\text{Eqn (C)}} \Rightarrow \tan \theta = 3$$

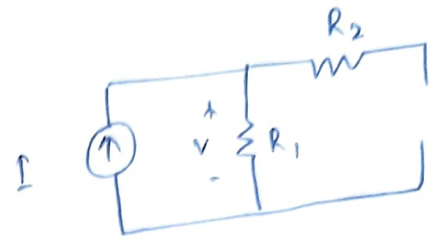
$$\boxed{\theta = 71.56^\circ}$$

$$\text{so, (B) } \Rightarrow A = -3.16$$

$$\text{so, (A) } \Rightarrow \boxed{i(t) = 3 - 3.16 e^{-2t} [\sin t + 71.56^\circ]}$$

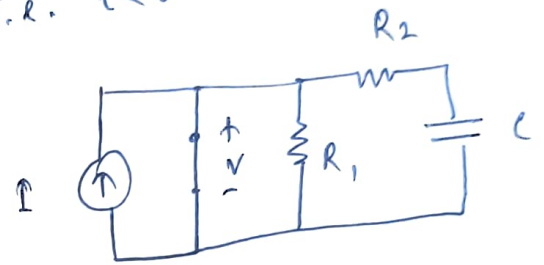
8. a) Draw the circuit in steady state once switch is opened at $t = 0$

$$V_{s.s.} = IR_1 \text{ V}$$



b) For natural response, draw the circuit in steady state when switch was closed i.e. $t < 0$

$$V(0) = 0 \text{ V}$$



c) complete response $V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$

$$\tau = (R_1 + R_2) C$$

so,
$$V(t) = IR_1 \left[1 - e^{-\frac{t}{(R_1 + R_2)C}} \right]$$